Research Development and Plan of Crash Analysis: Methods, Analysis and Future Directions

Final Project Report

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Declaration of Authenticity

I declare that this report: **"Predictive Analytics and Geospatial Analysis of Road Crash Patterns in Auckland"**, I have acted with integrity and ethics throughout this work. is my own work and that all the opinions, ideas and/or material of other people included in this project have been properly referenced, I have not submitted this report, or any part of it, for the award of any degree or professional qualification, or diploma or other similar title before in any other educational institution. I have acknowledged all assistance I have received in writing this report from whatever source. I affirm that I have used and cited such content in accordance with any applicable Terms of Service. Specifically:
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Abstract

Road Safety in New Zealand remains a problem. This master thesis deals with the problem of high severity crash location identification and crash prediction in Auckland based on ten years crash data publicly available from the NZTA Crash Analysis System (CAS, 2015-2025). Using a positivist, quantitative research design through a Microsoft Fabric Medallion Architecture (Bronze-Silver-Gold), this research four analytical streams: geospatial hotspot analysis, machine learning severity prediction, temporal trend analysis and vulnerable user risk profiling.

The Getis-Ord Gi* analysis for the gridsiles showed 36 statistically significant high convergence cells covering the entire corridor of SH1/Southern Motorway, having the Grafton–CBD zone at the 99.9% confidence level. An XGBoost classifier based on 111,657 crash records was trained, and it achieved an ROC-AUC of 0.7175 and a recall of 0.638, with lighting conditions and vulnerable user involvement as the two strongest severity predictors based on SHAP analysis. Holiday periods and crash severity were significantly associated as assessed by the χ-square test ($\chi^2 = 10.09$, $p = 0.018$) with Labour Weekend having the highest severity rate of 6.3%. Notably, crash involving pedestrian, cyclist, or motorcyclist were 8.95 times more likely to result in a fatal or serious injury than vehicle-only crashes (23.7% vs 2.6% severe rate).

Twelve evidence-informed recommendations are put forward, comprising targeted lighting improvements, specialised infrastructure for vulnerable users, Labour Weekend enforcement campaigns and ML-guided patrol routing, with an estimated combined annual benefit of NZD 15-20 million. The study critically reflects upon the ethical, cultural and methodological limitations it faced, including that crash hour information was not available in the public dataset, Te Tiriti o Waitangi responsibilities in data stewardship, and modifying standard ML practice in the New Zealand transport environment.

Introduction

New Zealand's Road to Zero action plan has an ambitious goal of decreasing road deaths and serious injuries by 40% from 2020 to 2030 (Ministry of Transport, 2019). Natural experiments: Auckland, New Zealand Despite substantial investment in infrastructure to improve road safety, Auckland, New Zealand's largest urban centre, still experiences disproportionately high crash rates, with over 111,000 crashes recorded in the Auckland region alone between 2015 and 2025 (NZTA, 2025). Around 4.8% of these are fatal or result in serious injury, and they place a heavy load on people, whānau, emergency services and the health system.

This capstone presents a critical comparison of the granularity of crash data available with that actually utilized in undertaking data-led safety work. Although the NZTA Crash Analysis System (CAS) holds more than a decade of georeferenced crash records, AT and road safety practitioners currently have very limited access to a suite of integrated spatial, predictive and temporal knowledge that could be used to support decisions on the allocation of scarce resources.

The report combines and continues from Assessment 1 (Research Proposal and Methodology) and Assessment 2 (Ethical and Cultural Analysis). This report details the analytical results, solution design, and the ethical and cultural measures incorporated across the study. Four questions were answered by the report: (RQ1) What are the locations of statistically significant clusters of high-risk crashes in Auckland? (RQ2) Is crash severity predictable by machine learning? (RQ3) Are there temporal (holiday) effects in crash severity? (RQ4) Which environmental and infrastructure predictors are most indicative of severe outcomes for vulnerable road users?

The research was carried out within Microsoft Fabric, leveraging a medallion lakehouse architecture, and applying the full NZTA CAS WGS84 dataset processing in Bronze ingestion, Silver cleaning and Gold feature engineering. All the analytical notebooks are versioned on GitHub and the results are presented in five pages Power BI dashboard targeting NZTA and Auckland Transport stakeholders.

Task 1: Final Research Report - Findings and Solution

1.1 Industry Problem and Context

The central industry challenge greatly addressed by this project is the fact that CAS data on high-severity crashes cannot be easily exploited for systemic identification, prediction and prioritization of high-severity crash locations in the Auckland Transport (AT) and New Zealand Transport Agency (NZTA) context, given the complex nature of the underlying data. Raw crash counts are publicly available, but these figures are seldom combined with spatial clustering analyses, machine learning predictions, or temporal risk analyses at a level of scale and detail applicable to operational decision-making.

There are many reasons why this problem is important. Firstly, New Zealand society bears an annual road fatality and serious injury social cost of more than NZ$ 4 billion, with each fatality estimated to cost NZ$ 4,840,000 and each serious injury NZ$ 662,000 (NZTA Social Cost Framework, 2024) . Second, raw crash counts are currently the primary basis for allocation of crash resources, which does not take into account the spatial clustering, predictive severity risk, or the outsized risk to vulnerable road users. Thirdly, the Road to Zero 40% reduction target needs such evidence-based intervention prioritization which existing reporting tools do not fully have.

This project is based on three well-recognized models from domain knowledge. The Safe System approach (NZTA, 2020) acknowledges that crashes are foreseeable and avoidable, with managing roads, speeds, vehicles, and road users in an integrated manner through a set of principles that directly informs the multi-dimensional aspect of this analysis. The CRISP-DM data mining process model (Chapman et al., 2000) organized the analysis workflow from business understanding to deployment. The design of the Power BI dashboard was based on the Nielsen usability heuristics (Nielsen, 1994) to make the results available also to the non-technical audience.

The project size is explicitly limited to crashes in the Auckland region recorded in CAS between January 2015 and December 2025, with road geometry, environmental conditions, crash place and crash result as variables. No driver data (alcohol, speed, licence status) is included in the project, as these are not available in the public CAS data set, a limitation that is crucial for understanding the upper bound for machine learning model performance.

Table 1: Summary of Industry Problem, Significance, and Domain Knowledge Alignment

Dimension	Detail	Domain Problem
Problem	No systematic data-driven crash severity prioritisation in Auckland	Safe System approach crash severity is preventable with data-driven interventions
Significance	NZD 4B+ annual social cost; Road to Zero -40% target by 2030	NZTA Social Cost Framework; Government policy obligation
Gap	CAS data available but not integrated for spatial + ML analysis	CRISP-DM process model applied to bridge raw data to actionable insight
Affected Parties	Pedestrians, cyclists, motorcyclists, NZTA, Auckland Transport, whānau of crash victims	Vulnerable user frameworks (WHO, 2022); Te Tiriti obligations to Māori disproportionately affected by road crashes

1.2 Literature Review Integration

A concise literature review allows four areas of the body of knowledge that are most applicable to the analysis used in this study and the extent of its findings to be recognized.

Geospatial Analysis of Road Crashes

Kernel Density Estimation (KDE) and Getis-Ord Gi* spatial statistics are the most used methods to detect crash concentration areas. Anderson (2009) shows that KDE is a suitable technique to smooth point event data to produce continuous crash density surfaces and that bandwidth selection based on Scott's rule of thumb gives an objective, data-driven method for parameter selection. Getis and Ord's (1992) Gi* statistic was introduced as a local spatial association measure, allowing us to classify grid cells into statistically significant hot spots at 95%, 99%, and 99.9% confidence levels, a multi-tier approach that the current analysis adapts in order to distinguish between extreme and moderate risk areas.

Spatial dependence was further confirmed by the global Moran's I (Moran, 1950) over the whole study area.

Summary statistics of Moran’s I provide evidence of significant positive spatial autocorrelation among fatal and severe crashes Similar significant positive spatial autocorrelation among fatal and severe crashes was found in this study, which lends support to the geospatial approach utilized in this study. The 1km × 1km grid with DistanceBand spatial weights (2km threshold), used in this study, is based on the methodology of Xie and Yan (2008), who proved that grid-based is a better option than point-based hotspot detection for big urban crash data.

Machine Learning Based Crash Severity Prediction

The use of ensemble techniques in predicting crash severity is also justified in the transport analytics literature. Compared to the Logistic Regression and Random Forest baselines, the XGBoost algorithm of Chen and Guestrin (2016) has consistently shown superior performance on tabular crash data, which can be mainly attributed to its treatment of diverse data types and imbalanced classes. However, Mannering et al. (2016) provide a pertinent critical insight for the current project there is a performance limit to crash severity models determined by data availability, with models that include road geometry and environmental attributes only generally obtaining ROC-AUC in the 0.65-0.70 range consistent with the results of this project.

SHAP (SHapely Additive exPlanations) values development Lundberg and Lee (2017) was motivated by the explainability limitation of black-box models. With individual feature attribution scores based on cooperative game theory, SHAP facilitates model validation (predictions are being driven by theoretically meaningful variables) and practical communication to non-technical stakeholders this is a stated need in the Assessment 1 learning outcome.

Temporal and Holiday Period Analysis

The temporal patterns literature on crash risk generally shows that holiday periods have higher crash risk, but the magnitude of that effect is substantially different between countries and type of holidays. Turner et al.(2012) examined New Zealand crash data and determined that the periods with the most severe crashes were the Easter and Labour Weekend periods, due to a mixture of factors including higher travel volumes, larger proportions of road trips on unfamiliar routes, and high levels of driver fatigue. The magnitude of the categorical association can be measured using Cramer's V effect size statistics, in relation to the nominal nature of the classification variables of crash severity and type of holiday.

Vulnerable Road User Risk

Globally, pedestrians, bicyclists, and motorcyclists are disproportionately impacted by fatal crashes. The physical risk mechanisms associated with road infrastructure (i.e., lack of body protection of road users, interaction with high-speed environments, insufficient infrastructure) are listed as three categories by WHO Global report on road safety (2023). Austroads (2021) identifies road lighting, road lane category and intersection type as the strongest modifiable risk elements for vulnerable users in the Australian and New Zealand context, a conclusion that is also support[ed) by the chi square and Cramér's V analysis conducted in this project. The 8.95× relative risk observed in this study is consistent with international data with Petzoldt et al. 2017) reported relative risks of 8-12 times for pedestrians in European urban areas.

1.3 Research Methodology

1.3.1 Research Design and Approach

This work adopts a positivistic research paradigm as it aligns with the quantitative and data-science related aspects of the industry problem. In a positivist framework, it is posited that objective patterns can be uncovered in crash data that can be quantitatively described through rigorous inquiry, and that knowledge obtained from such patterns is generalizable and can be utilized in directing future safety initiatives. This is warranted because the NZTA CAS data set is a census and not a sample of all reported crashes in New Zealand, thus allowing a large power for analysis.

The study uses a case study methodology with the aim of being both descriptive and predictive, with the case being limited to the region of Auckland, and time frame 2015-2025. The bordering decision was taken based on organizational relevance (Auckland Transport as the key stakeholder), data completeness, and computational feasibility within the Microsoft Fabric platform. The case study approach was chosen rather than a national study as case studies allow more thorough consideration of local spatial context (i.e., the SH1 Southern Motorway corridor).

The analysis pipeline is designed after the CRISP-DM process model tailored to lakehouse architecture, including: Business Understanding -> Data Understanding -> Data Preparation

-> Modelling -> Evaluation -> Deployment. Each phase corresponds to a specific notebook available in the project's GitHub repository.

1.3.2 Data Collection Methods

The main source of data is the **NZTA Crash Analysis System (CAS) WGS84** dataset, which was obtained from the NZTA Open Data Portal in December 2025. The dataset consists of 111,657 crash records in the Auckland region from January 2015 to December 2025, with 73 original columns related to crash location WGS84 (longitude/latitude), crash severity classification (Non Injury, Minor, Serious, Fatal), road geometry attributes (speed limit, number of lanes, road surface), environmental factors (weather, lighting, flatness), types of crash participant (pedestrian, cyclist, motorcycle, heavy vehicle), and financial year period.

No primary data were collected[surveys, interviews]. This was the rationale for Assessment 1 with respect to sufficiency of data the CAS dataset represents the entire population of reported crashes with sufficient specificity to answer all four research questions and ethical considerations, as approaching crash victims/families for primary data collection would necessitate further ethics approvals and could potentially cause distress.

Additional secondary data sources were NZTA Social Cost Framework (2024) for ROI calculation benchmarks and the Ministry of Transport Road to Zero strategy document for policy context.

1.3.3 Data Analysis Strategy

The analytical approach consisted of four parallel, but interrelated, streams of analysis, one for each research question:

• RQ1- Geospatial Analysis: KDE using scikit-learn KernelDensity with Scott's Rule bandwidth, followed by 1km grid Getis-Ord Gi* with the libpysal/esda modules. Global Moran's I also yielded supportive spatial autocorrelation results. This method was preferred to buffer zone analysis (conventional in NZTA methodology) because Gi* includes statistical confidence levels in its hot spot classification as opposed to arbitrary buffer distances.

• RQ2 - Machine Learning: Three classifiers were trained using a 80/20 stratified train/test split: Logistic Regression (baseline), Random Forest and XGBoost. Model selection was guided by class imbalance (4.8% severe crashes), with scale_pos_weight in XGBoost. SHAP TreeExplainer was run on the top model for feature attribution. MLflow monitored all the experiments in the Microsoft Fabric platform.

• RQ3 - Temporal analysis: Chi-square contingency analysis for holiday–severity relationship with Cramér's V effect size. Linear regression on yearly proportions of severe crashes was used to assess for trends. These approaches were chosen over time series analysis because no crash hour data was available in the dataset, and therefore, temporal resolution was restricted to year-by-holiday comparisons.

• RQ4 - Vulnerable User: Chi square and Cramér's V for ten potential risk variables and a binary vulnerable_user_involved indicator. Relative risk - the ratio of the severe crash rate in VU and non VU crashes was computed. Spatial references were also given for each use type, with folium heatmaps.

Table 2: Analytical Methods, Justification and Limitations

Method	RQ	Justification	Key Limitations
KDE + Gi*	RQ1	Statistical confidence levels for hotspot classification;	objective bandwidth selection Grid resolution (1km) may miss sub-kilometre clustering
XGBoost + SHAP	RQ2	Handles mixed types and class imbalance; interpretable via SHAP	No crash hour, driver speed, alcohol

			data limits AUC ceiling (~0.72)
Chi-square + V	RQ3	Appropriate for nominal categorical data; effect size quantifies practical significance	Temporal resolution limited to holiday period no crash hour in CAS
Relative Risk	RQ4	Directly interpretable by non-technical stakeholders actionable for intervention prioritisation	Confounding by road type not fully controlled in bivariate analysis

1.4 Findings and Analysis

1.4.1 Finding 1 - Geospatial Hotspot Identification (RQ1)

Conclusion: With the Getis-Ord Gi* statistics, there are 36 statistically significant grid cells (high-high clusters) along the SH1/Southern Motorway forming a long corridor, the Grafton-CBD zone again stands out at 99.9% confidence level ($Z > 3.29$, $p < 0.001$). Aggregate coverage for crashes across all hot spot tiers surpasses 50 %, meeting Objective 1.

The KDE was performed on severe crashes (fatal and serious injury) only, with a Gaussian kernel and Scott's rule bandwidth of about 0.024° (2.65 km at Auckland's latitude). A 200×200 grid (lon: 174.45-175.25, lat: -37.15 to -36.65) was computed over the real data extent. The 10 highest-density peaks were extracted with a minimum separation of 5km to avoid all peaks forming a spatial cluster in the maximum-density area.

Table 3: Top 10 KDE Hotspot Locations (Auckland Region, 2015-2025)

#	Road	Area	Crashes	Severe	KDE Score
1	KARANGAHAPE ROAD	Eden Terrace	1,497	71	Highest
2	MT WELLINGTON HWY	Mount Wellington	749	35	High
3	GREAT SOUTH ROAD	Clover Park	573	27	High
4	WEYMOUTH ROAD	Manurewa	544	26	High
5	NORTHERN MOTORWAY	Forrest Hill	355	17	Moderate-High
6	SOUTH-WESTERN MWY	Māngere	303	14	Moderate-High
7	LINCOLN ROAD	Henderson	275	13	Moderate
8	DOMINION ROAD	Mount Roskill	271	13	Moderate
9	GREAT NORTH ROAD	New Lynn	240	11	Moderate
10	CLEVEDON ROAD	SE Auckland	218	10	Moderate

The Gi* analysis provided a stratification between 36 hot spot cells. The Grafton CBD corridor (Eden Terrace/ Mt Wellington) was the only Hot Spot 99.9% achieving area, indicating an extreme and statistically significant High concentration of crashes. Hot Spot 99% classification was attained along the Southern Motorway (SH1) corridor at Manukau which encompassed 9 cells. Additional 14 cells were awarded with Hot Spot 95% classification, mostly in Henderson, New Lynn and Southeast Auckland.

Positive spatial autocorrelation was confirmed by Global Moran's I ($I > 0$, $p < 0.05$), indicating that severe crashes are spatially clustered rather than randomly distributed. This is a condition the meaningful use of Gi* local statistics. Critically, the kernel density estimation (KDE) and Gi* methods yielded cross-method replicated results, with both separately highlighting the Grafton CBD zone and SH1 Southern Motorway corridor as the highest concentrations of crash risk in Auckland, enhancing credibility in the spatial results. Nevertheless, a crucial methodological difference from the initial Assessment 1 guidelines was implemented: Gi* weights are calculated using total_crashes (all severities of crashes) as opposed to only severe_crashes. The correction was necessary since severe crashes make up only about ~4.8% of the total number of crashes, which resulted in the count for only the severe number becoming too sparse to obtain statistical significance in most of the grid cells. Applying "total crashes" as the weighting variable when restricting to severe crash coverage in the validation measures, generates a stronger spatial clustering signal while still directly addressing the question of interest.

1.4.2 Finding 2 - Machine Learning Severity Prediction (RQ2)

Summary Three classifiers were trained on the 111,657 crash records. XGBoost was chosen as the final model with ROC-AUC = 0.7175 and recall = 0.638 at threshold = 0.35. SHAP revealed lighting and vulnerable user involvement as top two predictors of crash severity. Using feature engineering, we derived 20 features which were ready for model training from the variables of road geometry, environmental conditions and crash context. To avoid leaking information, we did not use any of the variables indicating the number of injuries (fatalCount, seriousInjuryCount, minorInjuryCount) as features since they are consequences, not informative precursors, of severity. 2) Instead of SMOTE resampling (suitable for tree based models), we used the class imbalance parameter: scale_pos_weight (~20:1) in XGBoost.

Table 4: Model Comparison Results (XGBoost threshold=0.35)

Model	ROC-AUC	Recall	Precision	F1 Score	Vs LR
Logistic Regression (baseline)	0.731	0.582	0.120	0.199	-
Random Forest	0.712	0.427	0.160	0.233	-2.6%
XGBoost	0.718	0.638	0.083	0.147	-1.9%

That all three models converged in the range 0.71–0.73 AUC is analyzed as a success/finding, and not as a failure or discrepancy. This means that from the point of view of features, the dataset is already overfitted, more complexity of model will not improve prediction performance, because it is n that confounding factor, not model's capability. The publicly available CAS dataset does not contain variables of crash hour, driver speed, alcohol impairment, or vehicle condition that have been identified in the literature as dominant severity predictors (Mannering et al., 2016). A model ceiling around 0.72 AUC is in line with reported results for models with comparable features set.

XGBoost was chosen as the model to be used for deployment although it had slightly lower AUC than Logistic Regression (0.718 and 0.731 respectively) because it reaches considerably higher recall (0.638 versus 0.582) at the operational threshold of 0.35. For the operational use case of identifying outages most requiring intervention recall (sensitivity to severe outages) is the significance metric. A high-risk tag (prediction probability ≥ 0.35) was used to mark 38% of the test set crashes as requiring greater care.

The following is the ranked list of feature importance obtained by SHAP analysis (mean |SHAP| values): **lighting conditions (0.63) > vulnerable user involvement (0.59)** > poor visibility (0.38) > number of lanes (0.32) > intersection presence (0.31) > weather (0.30). The two highest SHAP features also independently validate the chi-square and relative risk results (Findings 1 and 4), offering cross-method verification. This convergence across independent analytical methods SHAP from ML, Cramér's V from chi-square is a major strength of the multi-stream research design.

Five-fold cross-validation results demonstrated that the XGBoost model generalizes with no overfitting (CV AUC standard deviation < 0.02) and fulfills robustness requirement for using as a patrol routing tool.

1.4.3 Finding 3 – Temporal and Holiday Patterns (RQ3)

Synopsis: Crash severity tends to be statistically significantly greater during holiday periods ($\chi^2 = 10.09$, $p = 0.018$), but the size of the effect is very small (Cramér's V = 0.0095). The severe crash rate is highest at Labour Weekend at 6.3% (+1.47 percentage points above an assumed baseline 4.8% for non-holiday crashes).

Summary results show a modest downward trend in the overall annual crash volume over 2015-2025, with dip in 2020-2021 due to lockdown restrictions resulting in reduced vehicle kilometres travelled (VKT). The proportion of fatal and serious crashes did not vary greatly throughout the decade, indicating systemic severity risk factors are likely to be persistent regardless of fluctuations in crash volume.

Table 5: Crash Severity Rates by Holiday Period

Period	Crashes	Severe %	Vs Baseline	Interpretation
Non-holiday (baseline)	~105,000	4.8%	-	Reference
Labour Weekend	~800	6.3%	+1.47pp	Highest severity period
Easter	~900	5.8%	+1.0pp	Elevated - rural routes
Christmas/New Year	~2,326	5.6%	+0.8pp	Highest volume period
Queen's Birthday	~600	5.2%	+0.4pp	Modest elevation

The small Cramér's V (0.0095) with the significant chi-square test ($p = 0.018$) demands a cautious interpretation. The statistical significance is a reflection of the large sample size rather than a strong effect in practical terms there are slightly higher rates of severity in the holiday periods, but the effect size is very small. This goes against any scenario where resources would be spread thin to cover all holiday periods, and suggests targeted enforcement on Labour Weekend, where the increase is most marked and where the operational response (state highway patrol on certain routes) is most clearly defined.

A In this regard, our study has a limitation related to the crash hour is not documented in the public NZTA CAS dataset. This restricts temporal analysis by comparing annual trends and holiday period trends. Both time-of-day and day-of-week patterns which are often the most useful temporal patterns for practitioners in international road safety work cannot be derived from these data alone. This limitation was acknowledged in the assessment 1 and is propagated to the dashboard where a mandatory note card informs the users of this constraint.

1.4.4 Finding 4 - Vulnerable User Risk Factors (RQ4)

Conclusions: Vulnerable road users (pedestrians, cyclists, motorcyclists) are 8.95 times more likely to sustain a fatal or serious injury in a crash than those involved in vehicle-only crashes. Ten environmental and infrastructure risk factors (Type of road lane, Type of road, Presence of roadside hazards, Number of lanes, Presence of road lighting, Lighting conditions, Use of a curved road, Use of a secondary road, Area type, level of precipitation) were found to be significantly associated with the involvement of a vulnerable user, and road lane type (Cramér's V = 0.125) and lighting conditions (V = 0.103) the two most modifiable factors.

Vulnerable user involvement was found in 11,458 (10.3% of the Auckland data set) crashes, including 4,598 pedestrian crashes, 2,510 cyclist crashes, and 4,573 motorcyclist crashes. The severe crash rate for vulnerable user crashes was 23.7%, compared with 2.6% for vehicle-only crashes - that is a relative risk of 8.95.

Table 6: Top 10 Risk Factors for Vulnerable User Crash Outcomes (Chi-Square Analysis)

Risk Factor	Chi-Square	p-value	Cramer's V	Implication
Road Lane Type	V = 0.125	<0.001	0.125	Off-road/1-way = highest risk
Lighting Conditions	V = 0.103	< 0.001	0.103	Dark unlit = highest risk
Urban Classification	V = 0.095	< 0.001	0.095	Open road = higher severity
State Highway	V = 0.087	< 0.001	0.087	SH routes = VU overexposure
Traffic Control	V = 0.073	< 0.001	0.073	Nil control = highest risk

Weather Conditions	V = 0.077	< 0.001	0.077	Heavy rain/mist elevated
Flat/Hill	V = 0.037	< 0.001	0.037	Hill = moderate elevation
Road Surface	V = 0.045	< 0.001	0.045	Unsealed = higher risk
Intersection	V = 0.062	< 0.001	0.062	Intersections = VU risk
Holiday Period	V = 0.018	0.041	0.018	Marginal significance
Flat/Hill	V = 0.037	< 0.001	0.037	Hill = moderate elevation

A comparison of crash locations of vulnerable road users also shows specific patterns by user type. Pedestrian crashes are clustered in the CBD and South Auckland town centres, aligning with areas of high pedestrian flow and pedestrian vehicle conflict points. Motorcyclist crashes exhibit the most extensive spatial coverage from the southern rural centre at Pukekohe, Clarks Beach, and the SH16 rural corridors which indicate that motorcycles are used for both urban commuting and recreational rural riding. Cyclist crashes are generally aligned with the CBD commuter corridor and regional recreational routes, but with two additional concentrations of bicycle crashes around shared paths.

Both the results of the pairwise models of travel distance and the relative-risk analysis in compartment-specific models have direct operational implications: action needs to be user-type specific. **This 8.95× relative risk finding is the single strongest quantitative finding of this research** and thus provides a strong evidence base to prioritise investments to improve vulnerable user safety in the suite of solution recommendations.

1.5 Solution Development

1.5.1 Proposed Solution and Recommendations

The proposed response is an integrated three-part road safety intelligence platform: a Microsoft Fabric analytics pipeline that generates four Gold Delta tables, a five-page Power BI dashboard for decision support to stakeholders, and a prioritized 12-recommendation action plan based on the findings. Each recommendation can be linked to a finding and contains a ROI estimate using the NZTA social cost figures.

Table 7: 12 Evidence-Based Recommendations with ROI Estimates

#	Recommendation	Basis	Priority	Est Annual ROI
R1	Install street lighting at Hot Spot 99.9% cells	Light V=0.103; dark conditions strongest severity predictor	CRITICAL	NZD 2.4M+
R2	Dedicated cycling/pedestrian infrastructure on off-road & 1-way routes	Road Lane V=0.125; highest modifiable risk factor	CRITICAL	NZD 3.2M+
R3	Upgrade traffic signals at top 20 Gi* hotspot intersections	Intersection significant p<0.001; nil control = highest risk	HIGH	NZD 1.8M+

R4	Targeted speed enforcement on open-road hotspots	Urban V=0.095; open roads = higher severity	HIGH	NZD 2.4M+
R5	Labour Weekend road safety campaign + police deployment	6.3% severe rate +1.47pp; SH routes highest risk	HIGH	NZD 800K+
R6	Motorcycle safety programme on state highway hotspots	SH V=0.087; motorcyclists on SH over-represented	HIGH	NZD 1.2M+
R7	Sealed surface maintenance on unsealed hotspot roads	Road surface p<0.001; unsealed = higher VU risk	MEDIUM	NZD 800K+
R8	Deploy XGBoost ML predictions for patrol routing	high_risk_flag cells; recall = 63.8% of severe crashes	MEDIUM	NZD 1.2M+
R9	Hill road safety audit for vulnerable users	flatHill V=0.037; motorcyclists on hill roads elevated	MEDIUM	NZD 800K+

R10	Real-time weather advisory at hotspot locations	weatherAV=0.077; heavy rain/mist significant	MEDIUM	NZD 400K+
R11	Pedestrian crossing upgrades at top 5 pedestrian hotspots	Pedestrian crashes 4.6% of all; 23.7% severe rate	MEDIUM	NZD 400K+
R12	Annual crash trend monitoring dashboard (Power BI)	Temporal trend slope automated alert if rate +5%/yr	ONGOING	Indirect

The total estimated annual benefit across the three streams (ROI streams) is **NZD 15-20** million, using NZTA Social Cost Framework figures: NZD 4,840,000 per fatality and NZD 662,000 per serious injury. Stream 1 (hotspot infrastructure, 5% crash reduction) is worth about NZD 8M; Stream 2 (vulnerable user safety, 10% severe crash reduction based on 8.95× relative risk) is worth about NZD 7M; and Stream 3 (Labour Weekend enforcement, 15% severe crash reduction) is worth about NZD 800K.

1.5.2 Implementation Considerations

The proposed recommendations for implementation require the involvement of several parties: NZTA (management of state highway infrastructure and national safety policy), Auckland Transport (local road operator with responsibility for public transport), Auckland Council (planning and land use), NZ Police (enforcement), as well as community groups for Māori and Pacific peoples (who are overrepresented in road crashes in South Auckland). Key constraints on implementation are: capital budgeting cycles (upgrades to infrastructure generally require 2-3 years of planning): procurement constraints for traffic signal and lighting upgrades, data governance constraints for introducing the ML patrol routing system in operational police work (Section 2 includes discussion of the ethical safeguards), and requirements for periodic data refresh due to annual updates of the CAS dataset.

The Power BI report visualisation is tailored for immediate use with minimal to no additional cost as it is built within the existing Fabric environment. The prioritisation of recommendations matrix (Table 7) is designed such that Auckland Transport can identify which interventions can be undertaken in the short, medium and long-term with the funding available for each of the five highest-severity hotspot cells.

1.6 Evaluation of Domain Knowledge Application

A critical reflection on the knowledge domain framework(s) used within this work points to aspects of good fit as well as challenge.

Safe System Framework (NZTA, 2020): The Safe System-based Safe System subculture proved to be an excellent guiding organizing framework, as it focused attention on the four pillars (safe roads, safe speeds, safe vehicles, safe road use) and required recommendations to be nature systemic rather than blaming the individual. However, the CAS dataset that we disclose to the public does not contain vehicle safety ratings and driver behaviour information, which means that it is not possible to directly investigate the “safe vehicles” and “safe road use” pillars. The SHAP feature importance has been adapted the same way with the logic that the fraction of the total feature importance that pertains over road versus driver context variables can be taken as an indicator of the relative importance of the two.

CRISP-DM Process Model: The CRISP-DM process is a good fit with the Microsoft Fabric stack, that uses the medallion architecture (Bronze -> Silver -> Gold) which aligns well with the phases of Data Understanding, Data Preparation, and Modeling. A limitation was the lack of formal Deployment phase of the ML model the predictions of the high_risk_flag are in

Gold layer but there is no production API or direct integration with Police. This is recognized as a next step.

Nielsen Usability Heuristics: In the Power BI dashboard design, all ten heuristics were considered. The most important changes for this domain where: (1) apply human-readable severity labels ('Fatal Crash', "Serious Crash') rather than database codes (severe_crash=1), complying with Nielsen's principle 2 (match between system and real world); and (2) make mandatory note cards on slides 3 and 4 outlining limitations of the data (AUC dataset limitations, no crash hour), complying with principle 10 (help and documentation). These modifications did not exist in generic Power BI report templates and needed to be made by design.

ML Model Performance Background: The biggest restriction on the application of the knowledge of the domain was the expectation (Assessment 1), that XGBoost would produce ROC-AUC > 0.85 and Recall > 0.80. These targets were not met (achieved: 0.718 and 0.638 respectively). As critical hindsight, these targets were based on published results for models developed from richer datasets including crash hour, speed and driver behaviour variables not available in CAS. The final model performance is in line with international benchmarks with comparable feature sets, and the 5-fold CV confirms generalisation. The restatement that the model obtains a meaningful predictive signal under the available data constraints is a much truer and more academically rigorous characterization.

2. Task 2: Ethical, Cultural and Risk Integration

2.1 Ethical Framework Integration and Domain Standards Applicability

Building on Assessment 2, higher-level ethical concerns for the design of domain standard were then formulated into three fundamental responsibilities data privacy and security, mitigation of algorithmic harm and responsible use of open government data. Here, each duty is critically reconsidered in the completed research, evidence of d-specific active agency efforts to meet these obligations is introduced, and the question of whether domain specific ethical norms and principles may have bent as a result of these agency efforts is entertained.

Data Privacy and Security

The NZTA CAS public dataset does not include any personally identifiable information crash records are aggregated at the grid-square level and contain no names, dates of birth, or vehicle registration. The privacy risk to individuals is substantially reduced. However, twolimit cases emerged in the study that warranted special attention.

Firstly, the georeferenced crash coordinates (WGS84 lat/long to six decimal places) can be considered a re-identification risk in rural areas with extremely small numbers of crashes per location. A single fatal crash at a given rural coordinate if combined with local knowledge could identify the victims. The risk was mitigated by: (a) conducting the analysis at the 1km grid cell level for hotspot analysis, not at the point-level, (b) excluding grids with <5 crashes from the public dashboard, and (c) keeping raw coordinate data only in the Gold layer, which is protected within the Fabric workspace.

Second, the high_risk_flag predictions of the ML model, when used for police patrol routing, are a kind of predictive policing that can be discriminatory. This risk is mitigated by the administrative controls described in Section 2.4.

Domain Standards: NZ Privacy Act 2020 and Data Governance

The New Zealand Privacy Act 2020 also contains IPPs that apply to the collection, use, and disclosure of personal information. Although the CAS dataset is not personal information as defined by the Act, we designed the project's processing to comply with IPP as best practices for responsible data governance for the benefit of the NZTA stakeholder audience.

Encryption at rest and in transit, role-based access control, and audit logging are some of the ways in which the Microsoft Fabric platform meets IPP 5 (storage security). IPP 11 (for processing non-disclosure outside of New Zealand) was treated as low risk, as Fabric data residency is aligned to be configured in Australia / NZ regions.

The 2023 guidance from the Office of the Privacy Commissioner on AI and automated decision making was referenced in the design of the safeguards for ML deployment. Guidance specifically recommends human oversight of automated decisions with significant impact on individuals, which directly informed the recommendation (R8) that ML predictions only be viewed as patrol routing decision support tools with human officer final decision authority, and not automated enforcement trigger.

Applicability and Limits of Domain Standards

Two separate domain standards needed to be modified for the context of New Zealand transport. The EU General Data Protection Regulation, Article 22 (right to explanation in relation to automated decisions) is not binding in New Zealand but was referred to in the development of the SHAP explanation dashboard confirming that model predictions can be explained to those who are on the receiving end of such predictions.

The ACRS data governance framework also advises that crash data should be processed only for road safety, not law enforcement or insurance. The project's guidance of using ML predictions to route patrols is right at the edge of this restriction. The adaptation in ML detection (which was again pointed out is at the location, not the individual) is that road safety outcomes, not individual enforcement always remain the focus.

2.2 Cultural Considerations

The assessment 2 analysis of ethical and cultural considerations highlighted that the Māori population was over-represented in Auckland road crashes and that South Auckland where the largest share of Māori and Pacific people live, contributes some of the worst crash rates across the region (including the Clover Park, Manurewa and Māngere hot spot areas identified in Finding 1).

Te Tiriti o Waitangi Obligations

Te Tiriti o WaitangiA foundational Crown-Māori relationship is articulated in te Tiriti o Waitangi that is now being threaded through data governance within the evolving framework of Māori Data Sovereignty (Tūhoe Aho Tākiri ki te Rangi, 2019). Three principles were followed during this project.

Tino Rangatiratanga (self-determination): We do not purport to speak for or represent Māori communities directly affected by road crashes. To invoke them is to present findings as technical 'facts' about publicly sourced information while explicitly allowing that communities have expertise in local crash contexts that this dataset does not reveal. Recommendations R1, R2 and R11 (lighting, cycling/pedestrian infrastructure, crossing upgrades) were also specifically prioritised in part because they respond to the geographic concentrations in South Auckland and in part because infrastructure investment is a direct community benefit rather than a surveillance measure. Partnership: The environment for intended release (Auckland Transport and NZTA) includes Crown agencies to which Treaty obligations attach. R12 (annual monitoring dashboard): The project's recommendation R12 (annual monitoring dashboard) notes that its indicators should be developed in partnership with tangata whenua so that the measures incorporated reflect community-defined concepts safety rather than technical optimization criteria alone."

Active Protection: The training data of the ML model is reflective of patterns in past crashes, which include the patterns of historical underinvestment in road safety infrastructure in South Auckland. As a result, the model risk predictions may underestimate severe crash risk in locations where crashes have been historically underreported or where infrastructure has lagged behind that in other locations. This limitation was documented in the model documentation and is noted in the Power BI dashboard help text.

Pacific Peoples and Culturally Responsive Design

Pacific communities also face over-representation in South Auckland crash data. The dashboard was developed to display results in plain language for non-technical audiences including local community partners, in alignment with culturally responsive communication practices. The severity rates are reported as absolute rates ("6.3 in every 100 crashes"), Not as a ratio to a national average which may not be relevant in the local community context.

2.3 Risk Review - New and Evolving Risks

Several ethical and cultural risks evolved or were newly identified during the research phase beyond those documented in Assessment 2.

Table 8: Ethical and Cultural Risk Register (Updated from Assessment 2)

Risk	Severity	Trigger/Context	Mitigation Applied
Predictive policing misuse	HIGH	ML patrol routing recommendation could be used for targeted enforcement against specific demographic groups in South Auckland	R8 framed as location-based patrol support only; human final decision required, Police oversight protocols required before deployment
Algorithmic bias - South Auckland	HIGH	Model trained on historically underinvested areas may systematically underestimate risk	Model documentation includes South Auckland bias note; dashboard includes data limitation card

Data re-identification (rural)	MEDIUM	Rural crash coordinates with unique context could re-identify individuals	Grid-level (1km) aggregation for public dashboard; cell suppression for n<5
Māori data sovereignty	MEDIUM	Crash data disproportionately reflects Māori communities	Community co-design recommended for R12 monitoring metrics; findings not attributed to cultural factors
Dashboard misinterpretation	LOW	Non-technical users may interpret AUC or Cramér's V incorrectly	Plain-language tooltips on all KPI cards; mandatory note cards for methodology limitations

2.4 Mitigation Strategy Update

The control strategies suggested in Assessment 2 have been updated and extended based on project finalization. Here are five specific updates.

Update 1 - ML Deployment Safeguards (new since A2): The XGBoost patrol routing model is subject to a formal risk assessment using an NZ Police-specific Algorithmic Impact Assessment framework (NZ Police, 2023). It needs to consider: Training data bias, Prediction accuracy disparities across demographics, Human in Oversight procedures, and Public

Disclosure obligations. This evaluation should be performed before being put into service and was not part of the research phase but a condition to carry out R8.

Update 2 - Suppression of Data Applied: A new cell suppression policy is now in use (j < 5 crashes are not disclosed to public on the dashboard) using a DAX filter in the Power BI dashboard data model. This provides a direct implementation of the privacy protection featured in Assessment 2, and is not part of the original Assessment 1 Specification.

Update 3 - Documenting Bias: Model documentation files in the GitHub repo now an explicit section entitled Known Biases and Limitations that discusses the South Auckland historical underinvestment context as well as the limitations in CAS data coverage (crashes not reported to Police are excluded from the data set, with under-reporting estimated to be more prevalent in communities possessing lower levels of trust in police engagement).

Update 4 - Community Engagement Recommendation Formalised: Recommendation R12 now clearly states that the monitoring measures for the annual dashboard review will be co-developed with Auckland Transport's Māori Responsiveness team and community members from the five hottestub areas. It did not appear in the mitigation measures for Assessment 2.

Update 5 - Cultural Sensitivity Check on Labels of Dashboard: All geographic labels presented on the dashboard are official suburb/locality and road names accessed from the CAS dataset, no colloquial naming is provided that could potentially be used pejoratively or stereotypically. The Southern Motorway corridor is defined by place and space, not by the identities of its border communities.

2.5 Reflections on the Broader Societal and Cultural Contexts

This project will inform New Zealand's Road to Zero approach by developing practical and evidence-based tools that can influence decisions around infrastructure investment and allocation of resources. The social benefits are likely to be huge: even if the suggested interventions succeed only a modest 5% of the projected return on investment (ROI), then the projected NZD 750K-1M annual gains in avoided crashes translate into real lives saved and injuries avoided, and these are mainly South Auckland communities most vulnerable to crash risk. Yet this project also lays bare the contradictions of [NCVS] oriented public security work. The same analytics infrastructure that has been used to support evidence-based road safety improvements including georeferenced crash records, ML predictions, spatial hotspot identification, can be coopted for surveillance, targeted enforcement, or insurance discrimination if used in ways are regulated. The project's contribution to ethical practice is not in the data or the models themselves it is in the explicit articulation of these risks and the governance protections needed to responsibly deploy. This reflects a sentiment articulated by the Algorithmic Accountability Aotearoa (2024) wherein responsible data science isn't just about technical correctness but rather who benefits and who is burdened by the outcomes of algorithmic systems.

Another unforeseen result mentioned during the project is the danger of focusing too much on historical crash records as a guide to future infrastructure. The CAS dataset reflects crashes that have happened in current infrastructure configurations it does not capture near-misses or the demand-suppressing effect of informal avoidance on the part of pedestrians and cyclists who steer clear of hazardous corridors. Recommendations focused on the highest-crash sites may unintentionally overlook potential high-risk sites where active modes are suppressed due to fear of conflict. This case could be made to support augmenting the crash data analysis by way of active travel demand modeling in the next iteration of the study.

Conclusion

This paper aimed to address how to analyze the decade-long Auckland crash data to produce meaningful road safety intelligence. It has been shown that the NZTA CAS dataset through a structured analytical toolset including geospatial statistics, machine learning, temporal analysis, and risk profiling of vulnerable road users, can lead to insights that are both statistically sound and actionable for transport agency stakeholders. The study's four major results are (1) that the SH1 / Southern Motorway corridor and Grafton CBD are Auckland's statistically verified crash hotspots, as independently confirmed by KDE and Gi* at the 99.9% significance level (2) XGBoost produces non-trivial crash severity prediction (ROC-AUC 0.718, recall 0.638) based on road environment attributes only, lighting and vulnerable user involvement as the most influential factors a finding that accords with international literature under similar data limitations (3) holidays are statistically but not practically significantly related to higher severity, Labour Weekend being the practical actionable exception, and (4) **vulnerable road users are at a 8.95× elevated relative risk of fatal or serious injury** with road lane type and lighting being the most controlable risk factors. Twelve recommendations based on evidence are put forward with an overall annual ROI estimate of NZD 15-20 million. Critical reflection on the research process makes evident three significant methodological modifications applied due to real data limitations (a) the Gi* weight variable was adjusted from severe crashes to total crashes due to sparsity (b) ML performance goals as outlined in Assessment 1 were not attained as crash hour and driver behaviour variables were missing, this limitation is now revealed and contextualized instead of occluded and (c) the temporal analysis was reduced from anticipated hourly sequences to holiday status comparisons, as crash hour is not available in the public dataset. From an ethical perspective, the project wove data governance safeguards throughout: cell suppression for low-count geographies, human review for ML patrol routing, bias reporting for South Auckland, and Te Tiriti o Waitangi-informed suggestions for community-based design of monitoring indicators. These protections are not tangential to the research they are built into the design of the intervention itself. The next stage for this work will be: integration with real-time crash notification API from NZTA for dataset refresh automation; a formal Algorithmic Impact Assessment to be completed prior to the deployment of ML patrol routing; co-design of monitoring metrics with tangata whenua and Pacific community representatives; and the analysis to be expanded to incorporate near-miss incident reports and active travel demand data to redress the crash record's under-representation of suppressed mode trips.

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Appendices

Appendix A - Microsoft Medallion Architecture

The data pipelining is based on a four-layer medallion architecture in Microsoft Fabric:

Bronze Layer: Raw NZTA CAS WGS84 data loaded into Delta tables using PySpark. Maintains original 73 columns. No filtering or modification.

Silver Layer: Data cleaning performed–null imputation, severity classification normalisation, coordinate validation (WGS84 bounds check), and elimination of records outside Auckland boundary.

Gold Layer: Feature engineering that will output 111,657 clean crash records with 40+ derived features such as distance_from_cbd_km, is_intersection, vulnerable_user_involved, and severity binary target (severe_crash).

Dashboard Layer: This solution layer includes nine Gold Delta tables imported into Power BI Desktop (Export to CSV [no Fabric Premium capacity] is connected).

Appendix B - Machine Learning Feature List

Final feature set used in XGBoost model

Numeric: speedLimit, NumberOfLanes, distance_from_cbd_km, is_cbd_area, has_pedestrian, has_bicycle, has_motorcycle, vulnerable_user_involved, is_state_highway, poor_visibility, is_intersection, complex_traffic_control

Categorical (label-encoded): weatherA, light, roadSurface, trafficControl, flatHill_clean, roadLane_clean, urban_clean, roadCharacter

Appendix C - Power BI Dashboard Structure

Five-page dashboard deployed to Microsoft Fabric workspace:

Page 1 - Executive Overview: KPIs (Total Crashes, Fatal Crashes, Severity Rate, Best ML AUC); annual trend line chart; severity donut; top 10 roads bar chart

Page 2 - Geospatial Intelligence: Gi* hotspot map, Hotspot Coverage % and Severe Coverage % KPIs, bar chart by classification tier, top 20 grid cells table

Page 3 - Predictive Insights: Model comparison table, ROC-AUC bar chart, high-risk crash map; SHAP feature importance image

Page 4 - Temporal Analytics: Annual area chart; holiday severity bar chart, Labour Weekend highlight, mandatory data limitation note cards

Page 5 - Vulnerable User Safety: 8.95× Relative Risk KPI (most prominent), VU Severe Rate (23.7%) and Non-VU Rate (2.6%) KPIs, 100% stacked severity bar, top 10 risk factors by Cramér's V

Appendix D - Folder Structure

notebooks/ - 00 through 09: setup, bronze ingestion, data dictionary, silver cleaning, gold features, geospatial analysis, ML models, temporal analysis, vulnerable user analysis, validation

docs/ - architecture_diagram.png

reports/ - model_comparison.csv, hotspot_results.csv, recommendations_roi.csv, shap_importance_bar.png

outputs/ - kde_heatmap_top10.html, hotspot_map_gistar.html, hotspot_pedestrian.html, hotspot_cyclist.html, hotspot_motorcyclist.html